Part 2

Advanced generative AI techniques and deployment

With the basics of building full-stack generative AI web applications with Next.js and the Vercel AI SDK under your belt, it's time to go a step further. In this part, we'll dive deep into advanced techniques and the critical steps needed to take your applications from concept to production. You've already laid the foundation; now, let's make something truly remarkable and get it ready for the real world.

We'll begin with the principles of prompt engineering, showing you how to craft effective prompts to get the exact outputs you need from large language models. Next, you'll be introduced to LangChain.js, a powerful framework that simplifies building complex, context-aware AI applications, with a focus on advanced use cases. Building is only half the battle, so we'll then cover the crucial stages of testing and debugging your applications to ensure they perform as expected. To wrap it up, I'll guide you through the essential steps for securely deploying your applications, making them accessible and reliable for your users.